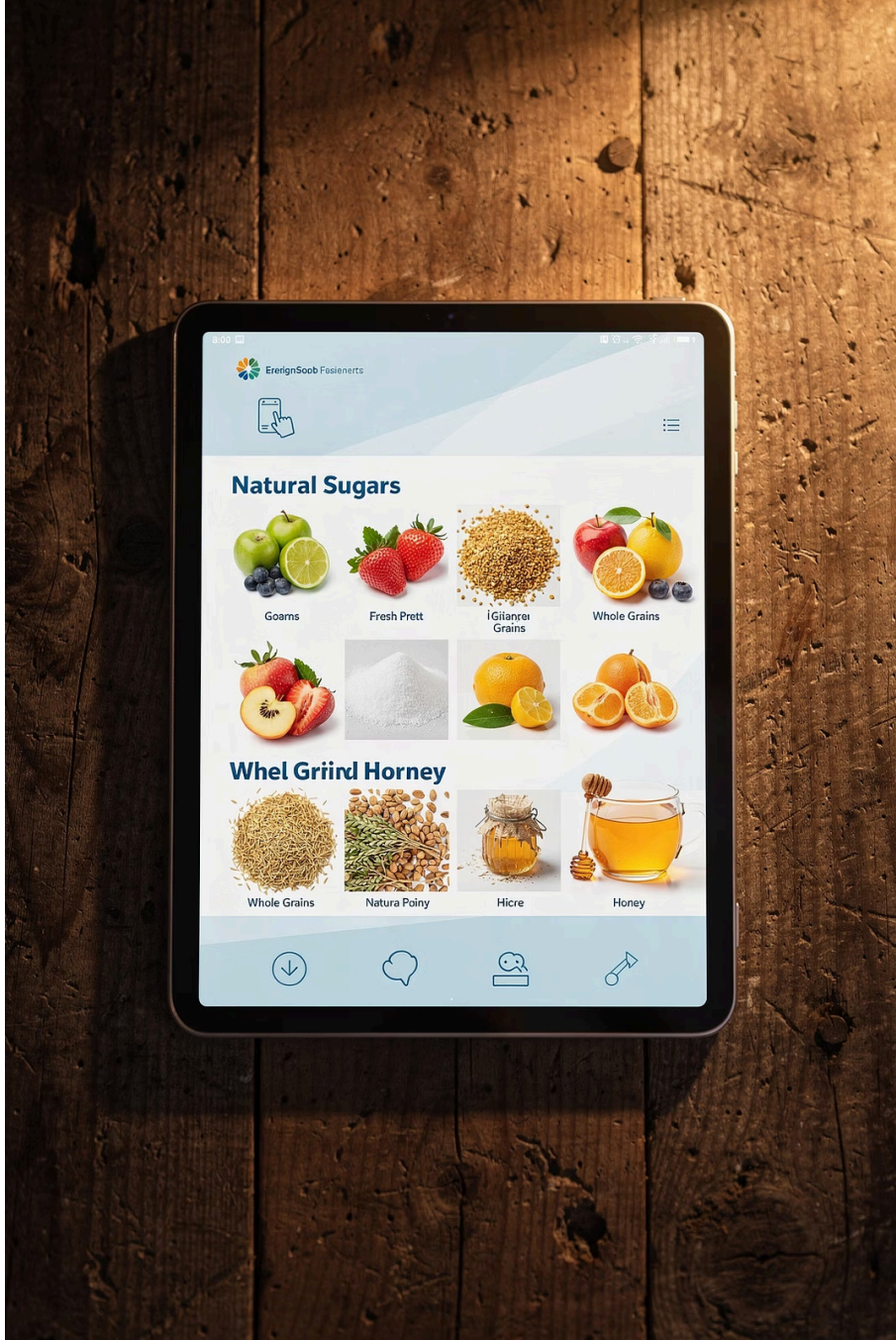
AP BIOLOGY • UNIT 1

TOPIC 1.4

Carbohydrates: Structure & Function

From simple sugars to complex polymers — understanding how carbohydrates are built, connected, and used by living organisms.

Learning Objective: Describe the structure and function of carbohydrates.



📋 OVERVIEW

What We'll Cover

This study set walks through the essential knowledge for Topic 1.4 — from the smallest sugar molecules to the massive polysaccharides that store energy and provide structural support in living systems. Use these slides as a self-study review or classroom reference.

01

Monosaccharides

The monomers — simple sugars like glucose, fructose, and galactose that serve as building blocks.

02

Polymers & Bonding

How covalent bonds (specifically glycosidic linkages) connect monomers into polysaccharides.

03

Linear vs. Branched

Why the shape of a polymer matters — and how it determines function in cells.

04

Key Examples

Cellulose, starch, and glycogen — the three polysaccharides you need to know for the AP Exam.

05

Review & Practice

Summary comparisons, key vocabulary, and exam-style thinking to lock in your understanding.

MONOMERS

Monosaccharides: The Simple Sugars

Monosaccharides are the **monomers** of carbohydrates — the smallest, simplest sugar units that cannot be broken down into smaller sugars by hydrolysis. They are the fundamental building blocks from which all larger carbohydrates are constructed. Understanding monosaccharides is the first step to understanding the entire carbohydrate family.

Key Characteristics

- **General formula:** $(CH_2O)n$, where n is typically 3–7. This means carbohydrates are literally "hydrates of carbon" — carbon atoms bonded to water molecules.
- **Soluble in water** due to their many hydroxyl (–OH) groups, which form hydrogen bonds with water molecules.
- **Sweet-tasting** — the sweetness you perceive in fruits and honey comes directly from monosaccharides.
- **Quick energy source:** Cells can rapidly break down monosaccharides during cellular respiration to produce ATP.
- Most biologically important monosaccharides are **hexoses** (6 carbons), though pentoses (5 carbons) like ribose are also critical (think: RNA and DNA).

Common Examples

Glucose

The most important monosaccharide in biology. It is the primary fuel for cellular respiration and the monomer used to build starch, glycogen, and cellulose.

Fructose

Found abundantly in fruits and honey. It is an isomer of glucose (same formula $C_6H_{12}O_6$, different structure) and is the sweetest natural sugar.

Galactose

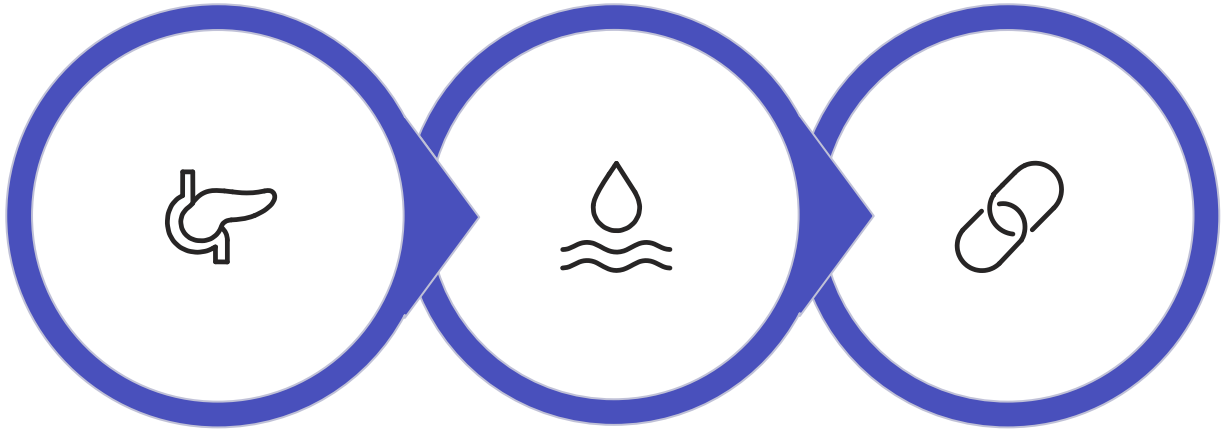
A component of lactose (milk sugar). Also an isomer of glucose, differing only in the arrangement of atoms around one carbon.

AP Exam Tip: You do NOT need to memorize the specific ring structures of individual monosaccharides. Focus instead on the concept that monosaccharides are monomers, they share the general formula $(CH_2O)_n$, and glucose is the most biologically significant example.

BIG IDEA

Monomer → Polymer: How Carbohydrates Are Built

One of the most fundamental concepts in AP Biology is the **monomer-to-polymer relationship**. Just as letters combine to form words and sentences, monosaccharides link together to form disaccharides and polysaccharides. This process is central to how organisms store energy and build structural components.



Monosaccharides → **Dehydration Synthesis** → **Polysaccharide**

The key reaction is **dehydration synthesis** (also called a condensation reaction). During this process, a covalent bond — specifically a **glycosidic linkage** — forms between two monosaccharides, and a molecule of water (H_2O) is released as a byproduct. The reverse reaction, **hydrolysis**, breaks these bonds by adding water back, releasing individual monomers. This is how your digestive system breaks down the starch in bread back into glucose molecules your cells can use.

🔗 Dehydration Synthesis

- Removes a water molecule (–H from one monomer, –OH from another)
- Creates a **covalent glycosidic bond**
- Builds larger molecules (anabolic process)
- Requires energy input (enzyme-catalyzed in cells)

💧 Hydrolysis

- Adds a water molecule to break the bond
- Breaks the **glycosidic bond** apart
- Releases monomers (catabolic process)
- Releases energy; used during digestion

Key Vocabulary: A **glycosidic linkage** is the specific type of covalent bond that connects monosaccharides. You'll see this term repeatedly — it's the "glue" that holds all polysaccharides together.

POLYMER SHAPE

Linear vs. Branched Polysaccharides

When monosaccharides polymerize, the resulting polysaccharide can take on different shapes depending on **how and where the glycosidic bonds form**. This structural difference has profound consequences for function. The shape of the polymer directly determines whether it serves as an energy storage molecule or a structural component.

Linear (Unbranched) Chains

Monomers are connected end-to-end in a single, straight (or gently curved) chain. These chains can align side by side and form strong, rigid structures through hydrogen bonding between adjacent chains. **Example: Cellulose** — the rigid cell walls of plants are built from linear chains of glucose bundled tightly together.

Branched Chains

In addition to the main chain, side branches extend off at various points. Branching creates a more compact, globular shape and — critically — provides many "ends" from which enzymes can simultaneously clip off glucose monomers. This makes branched polysaccharides ideal for **rapid energy mobilization**. **Example: Glycogen** — highly branched, allowing animals to quickly access stored glucose.

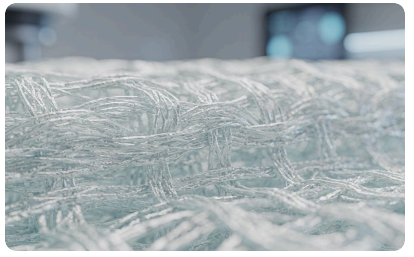
Why Does Shape Matter?

This is a recurring theme throughout AP Biology: **structure determines function**. A linear polysaccharide like cellulose forms tough fibers because the chains pack together via hydrogen bonds. A branched polysaccharide like glycogen stores energy efficiently because its many branch endpoints allow rapid enzymatic breakdown. Even starch comes in two forms — amylose (linear, helical) and amylopectin (branched) — each with slightly different properties. The degree of branching is one of the key variables that nature uses to fine-tune carbohydrate function.

CHAPTER 2 ILLUSTRATIVE EXAMPLES

The Three Polysaccharides You Need to Know

The AP Biology curriculum highlights three polysaccharides as illustrative examples of how the same monomer — **glucose** — can be assembled into polymers with dramatically different structures and functions. All three are made entirely of glucose, yet they behave very differently in living organisms because of differences in bonding and branching.



Cellulose

Found in: Plants (cell walls)
Structure: Linear, unbranched chains
Function: Structural support

Cellulose is the most abundant organic molecule on Earth. Its linear glucose chains are held together by hydrogen bonds, forming rigid microfibrils that give plant cell walls their strength.

Starch

Found in: Plants (roots, seeds, tubers)
Structure: Mix of linear (amylose) and branched (amylopectin)
Function: Energy storage

Plants store excess glucose as starch in plastids. Starch granules can be broken down by hydrolysis to release glucose when the plant needs energy, especially during nighttime or dormancy.

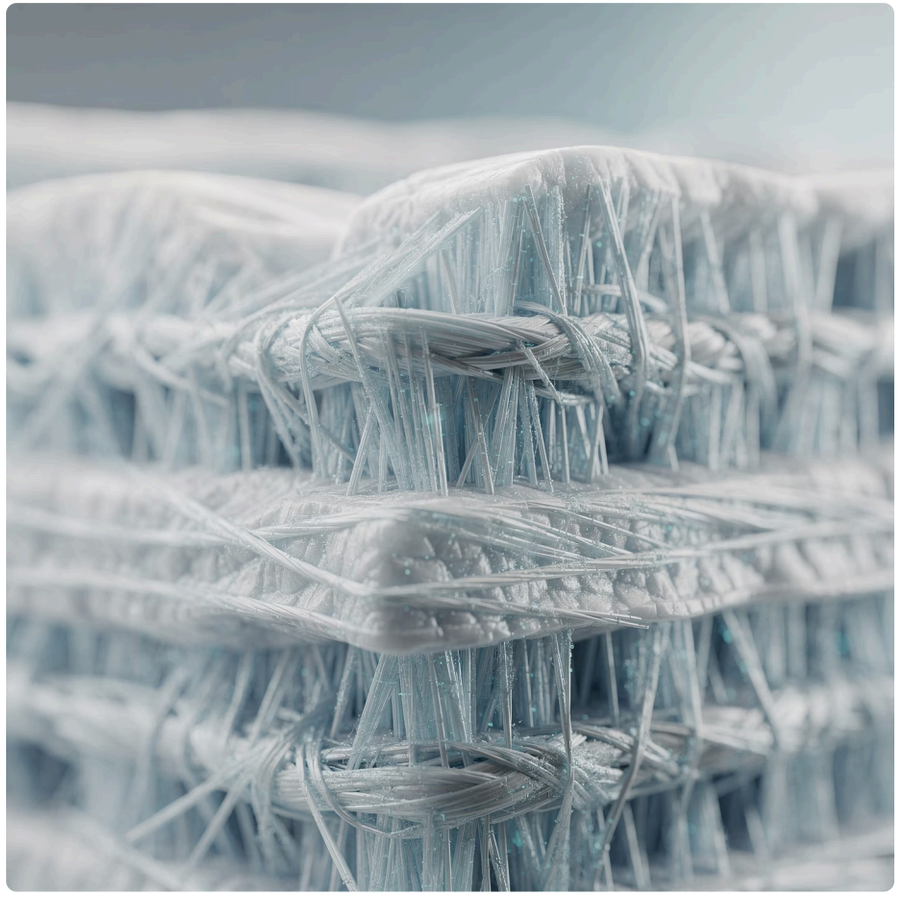
Glycogen

Found in: Animals (liver, muscle cells)
Structure: Highly branched
Function: Short-term energy storage

Glycogen is the animal equivalent of starch. Its extensive branching allows rapid glucose release — critical during exercise or between meals when blood sugar drops.

DEEP DIVE

Cellulose: Nature's Structural Masterpiece



Structure

Cellulose is composed of **long, unbranched chains of glucose** monomers linked by β-1,4 glycosidic bonds. (You don't need to memorize the bond type for the AP Exam, but understanding that the bond orientation differs from starch helps explain why cellulose behaves differently.) These linear chains run parallel to one another and are cross-linked by **hydrogen bonds**, forming bundles called **microfibrils**. The microfibrils are incredibly strong — comparable to steel on a per-weight basis.

Function

- Provides **rigid structural support** for plant cell walls, allowing plants to grow upright and resist mechanical stress.
- Acts as a protective barrier around each plant cell.
- Is **indigestible by most animals** — humans lack the enzyme (cellulase) needed to break β-glycosidic bonds. This is why cellulose passes through our digestive system as dietary fiber.
- Some organisms (termites, ruminant animals like cows) can digest cellulose thanks to symbiotic microorganisms in their guts that produce cellulase.

Fun Fact: Cellulose makes up roughly 33% of all plant matter. Cotton is nearly pure cellulose (~90%), and paper is essentially processed cellulose. Every time you write on a piece of paper, you're interacting with a glucose polymer!

DEEP DIVE

Starch: The Plant's Energy Reserve

Starch is how plants store surplus glucose produced during photosynthesis. When a plant makes more glucose than it needs immediately, it polymerizes the excess into starch and stores it in specialized organelles called **amyloplasts** (a type of plastid). When the plant later needs energy — such as at night when photosynthesis stops, or during germination of a seed — it hydrolyzes the starch back into glucose.

Amylose

The **linear** component of starch. Amylose consists of unbranched chains of glucose that coil into a helical (spiral) shape. It typically makes up about 20–30% of starch. Because it is linear, amylose is more compact and slightly harder for enzymes to access, meaning it is digested more slowly.

Amylopectin

The **branched** component of starch. Amylopectin has a branch point roughly every 24–30 glucose units, creating a bushy, tree-like structure. It makes up about 70–80% of starch. The branches provide multiple sites for enzyme action, so amylopectin is digested more rapidly than amylose.

Where Do We Find Starch?

Plants concentrate starch in **roots, tubers, seeds, and fruits** — essentially, anywhere the plant needs to store energy for later use. Potatoes, rice, wheat, corn, and cassava are all starch-rich foods that form the caloric backbone of human diets worldwide. When you eat these foods, your salivary and pancreatic amylase enzymes hydrolyze starch into maltose and eventually glucose, fueling your cells.

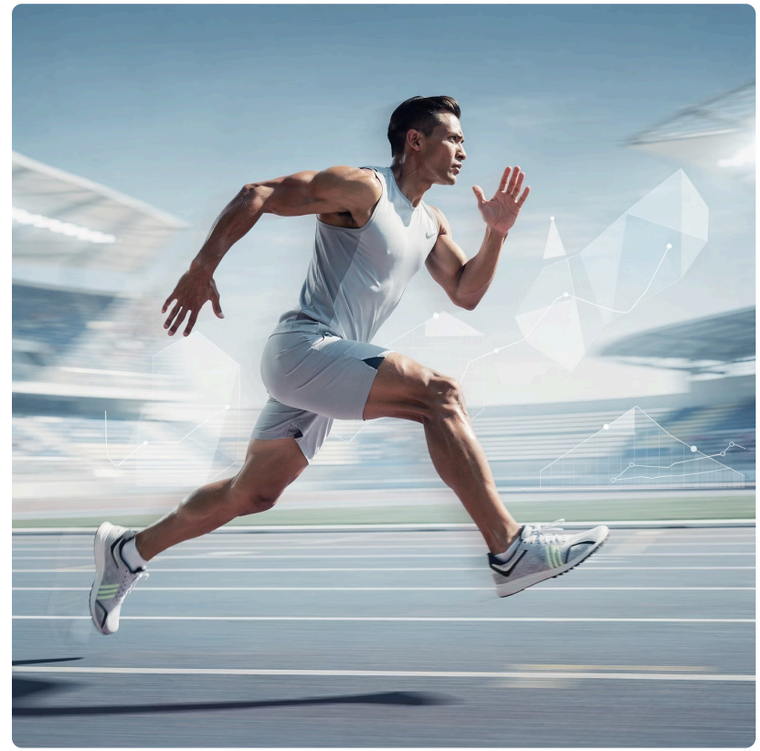
DEEP DIVE

Glycogen: The Animal Energy Reserve

Glycogen is often described as "animal starch" because it serves the same basic function as starch in plants — **storing glucose for later use**. However, glycogen is structurally distinct: it is **far more branched** than amylopectin, with branch points occurring every 8–12 glucose units compared to every 24–30 in amylopectin. This extreme branching is not accidental — it's an adaptation for the energy demands of animal life.

Why So Branched?

Animals need to mobilize glucose **quickly and in large amounts** — think of a sprinting cheetah or a fleeing gazelle. The highly branched structure of glycogen means there are **many terminal glucose residues** available simultaneously for enzymatic cleavage. Multiple enzyme molecules (glycogen phosphorylase) can work on different branches at the same time, producing a burst of glucose release far faster than would be possible from a linear chain.



Where Is Glycogen Stored?

Liver (~100 g)

Maintains blood glucose levels between meals. Liver glycogen can be broken down and exported as glucose to the bloodstream for use by the entire body.

Skeletal Muscle (~400 g)

Fuels muscle contraction during exercise. Muscle glycogen is used locally — it is not shared with other tissues.

AP Connection: Glycogen stores in the human body are limited (enough for roughly 24 hours of moderate activity). Once glycogen is depleted, the body switches to breaking down fats and eventually proteins — which is why marathon runners "hit the wall" and why carbohydrate loading before endurance events is a common strategy.

COMPARISON

Cellulose vs. Starch vs. Glycogen: Side by Side

All three polysaccharides are polymers of glucose, yet their different bonding patterns and branching structures give them vastly different properties and functions. This table summarizes the key distinctions you should know for the AP Exam.

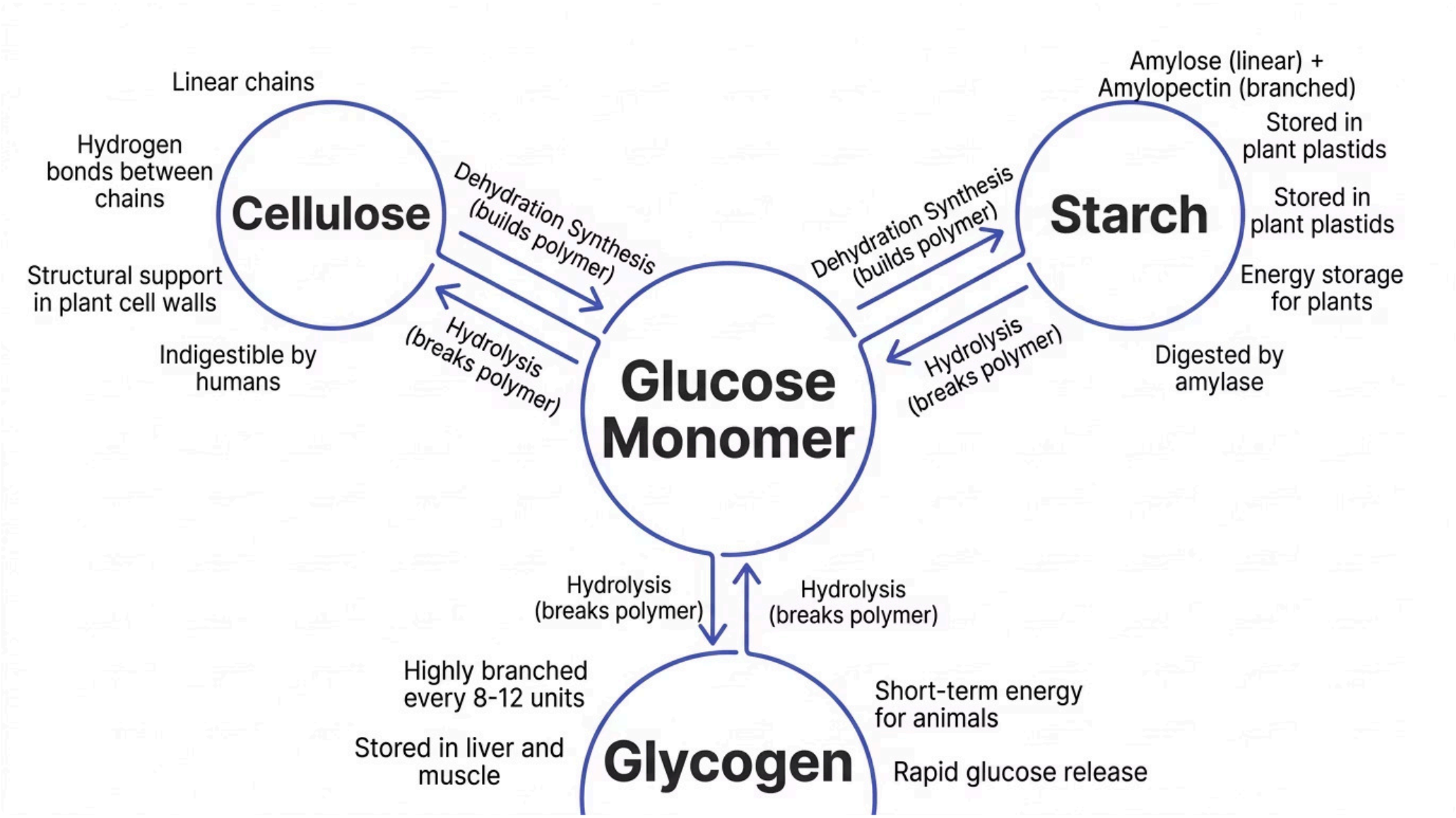
Feature	Cellulose	Starch	Glycogen
Monomer	Glucose	Glucose	Glucose
Found in	Plants (cell walls)	Plants (roots, seeds, tubers)	Animals (liver, muscle)
Primary Function	Structural support	Energy storage	Short-term energy storage
Structure	Linear, unbranched	Amylose (linear) + Amylopectin (branched)	Highly branched
Branching	None	Moderate (amylopectin)	Extensive (every 8–12 units)
Digestible by humans?	No (dietary fiber)	Yes (amylase enzyme)	Yes (phosphorylase enzyme)
Key Adaptation	H-bonds between chains → rigidity	Compact storage in plastids	Many branch ends → rapid glucose release

Remember the pattern: Same monomer (glucose) → different bonding arrangements → different polymer shapes → different functions. This is a perfect example of the AP Biology theme: **structure determines function**.

VISUAL SUMMARY

From Monomer to Function: The Big Picture

This infographic captures the entire story of carbohydrate structure and function in one view. Trace the path from a single glucose monomer through polymerization to the three major polysaccharides and their roles in living organisms.



Notice how all three polysaccharides originate from the same monomer but diverge based on bond type and branching pattern. The **dehydration synthesis** reaction builds the polymers; **hydrolysis** breaks them back down. This reversibility is what makes carbohydrates such effective energy currencies in biological systems.

EXAM PREP

Key Takeaways for the AP Exam

1 Monomers & Polymers

Monosaccharides (e.g., glucose) are the monomers of carbohydrates. They are linked together by **covalent glycosidic bonds** through dehydration synthesis to form **polysaccharides** (polymers). Hydrolysis reverses this reaction.

2 Structure Determines Function

The way monomers are bonded — and whether the resulting polymer is **linear or branched** — directly determines the polysaccharide's biological role. Linear → structural (cellulose). Branched → energy storage (starch, glycogen).

3 Three Key Examples

Cellulose = structural support in plant cell walls (linear). **Starch** = energy storage in plants (linear + branched). **Glycogen** = short-term energy storage in animals (highly branched for rapid access).

4 What's NOT on the Exam

You do **not** need to know the specific molecular structures of carbohydrate polymers (e.g., α vs. β linkages, ring structures). Focus on the conceptual relationship between monomer, polymer, bonding, shape, and function.

Bottom Line: If you can explain how glucose monomers are joined by covalent bonds to form polysaccharides, and why the linear vs. branched structure of cellulose, starch, and glycogen leads to different biological functions, you've mastered Topic 1.4.